

DIGITAL SYSTEMS AND COMPUTER ARCHITECTURE

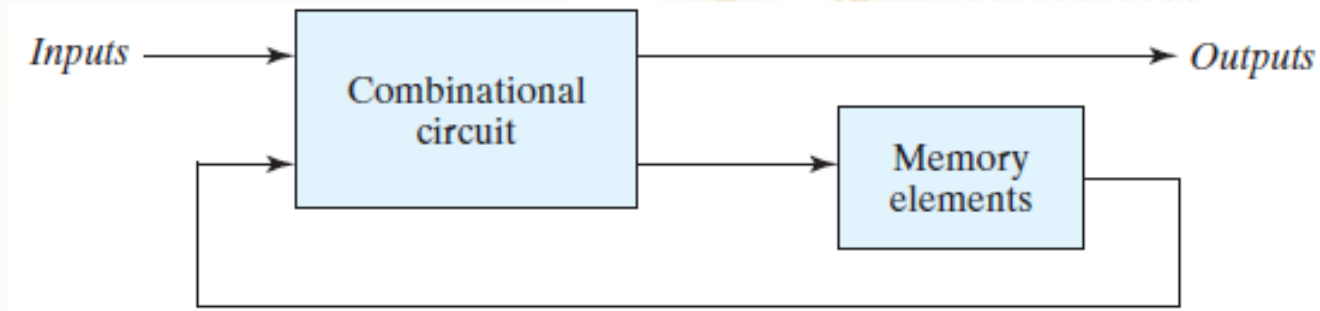
Unit 3: Sequential Circuits

Session 3.1: Focus

- Sequential Circuits
- Classification of Sequential Circuits
 - Asynchronous
 - Synchronous
- Storage Elements
- Level & Edge Triggered
- Latch & Flipflop
- SR Flipflop

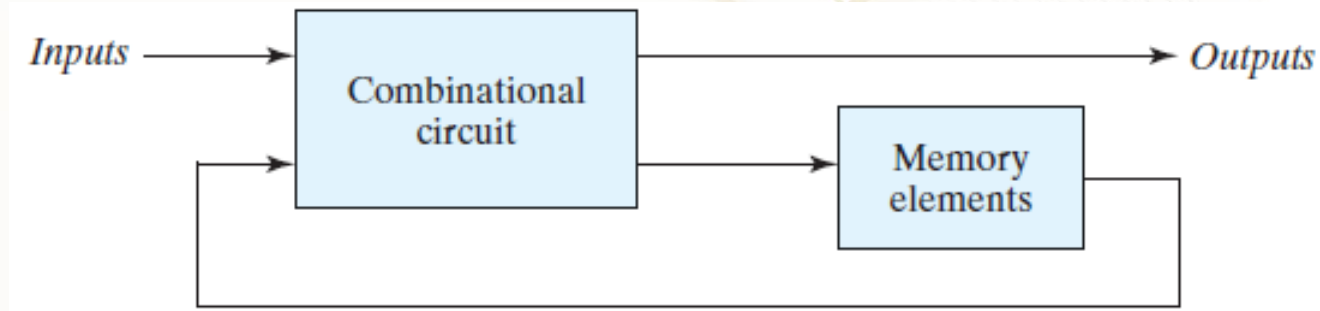
Sequential Circuits

Sequential Circuits



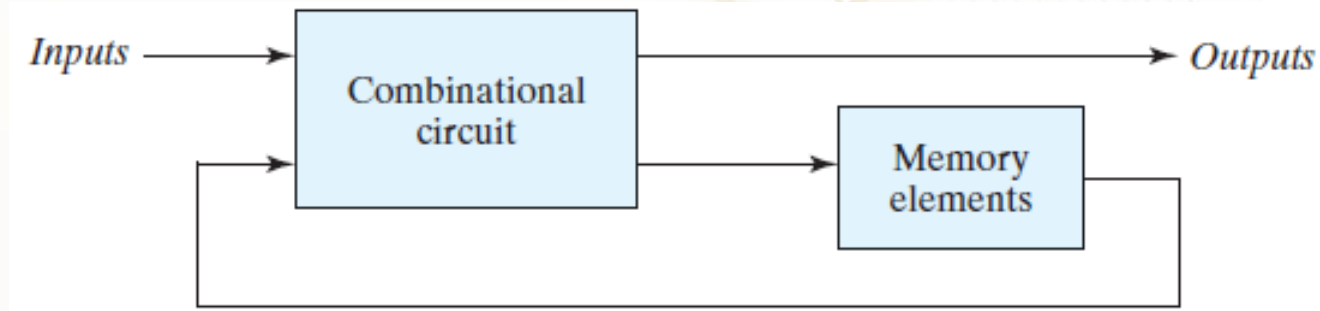
- **Combinational circuits** with **feedback** elements which store the internal **states** (0 or 1) of the circuit are called **Sequential circuits**.
- The **outputs** of **Sequential circuits** **depend** not only on the **inputs** but also on the **internal stored states** of the circuit
 - Based on the types of storage elements used to store the state, the sequential circuits can be classified

Sequential Circuits ... contd.



- **Binary information stored** in these elements at any given time **defines the state of the sequential circuit** at that time.
- The sequential circuit **receives** binary information from **external inputs** that together with the present state of the storage elements,
 - determine the binary **value** of the **outputs**.

Sequential Circuits ... contd.



- The next state of the storage elements is also a function of external inputs and the present state.
- Thus, a **sequential circuit** is specified by a time sequence of **inputs**, **outputs**, and **internal states**.
- In contrast, the **outputs** of **combinational logic** depend only on the **present values** of the **inputs**.

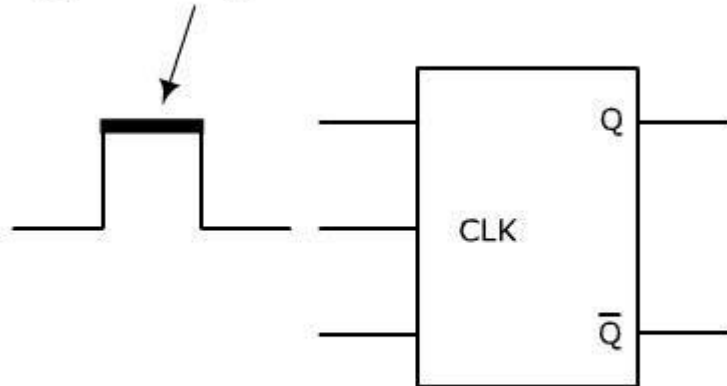
Classification of Sequential Circuits

Classification of Sequential Circuits

- There are **two** main **types** of sequential circuits, and their classification is
 - A function of the **timing** of their **signals**
- **Synchronous sequential circuit**
- **Asynchronous sequential circuit**

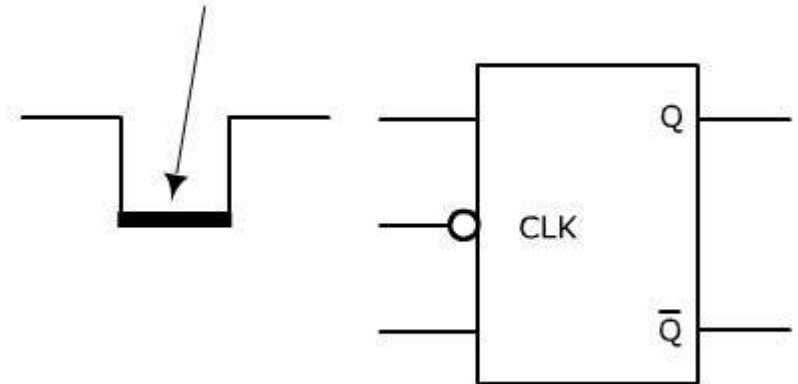
Level and Edge Triggering of Flip-flops

Triggers on high clock level



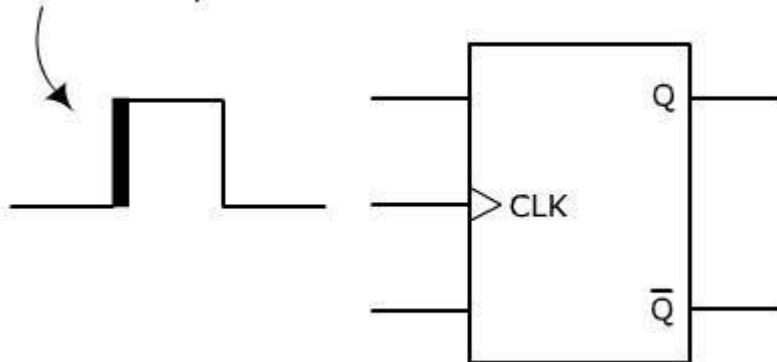
High Level Triggering

Triggers on low clock level



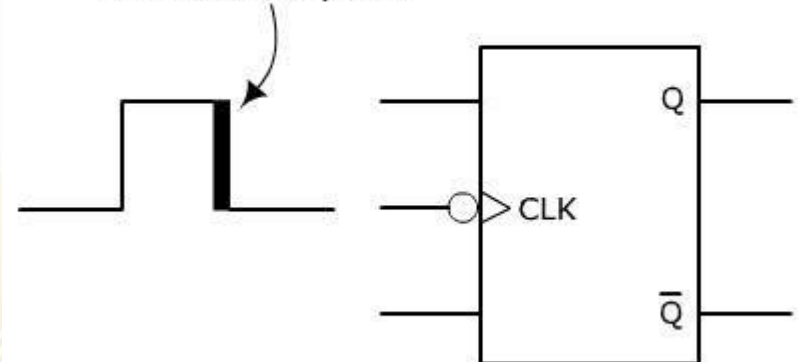
Low Level Triggering

Triggers on this edge
of the clock pulse



Positive Edge Triggering

Triggers on this edge
of the clock pulse

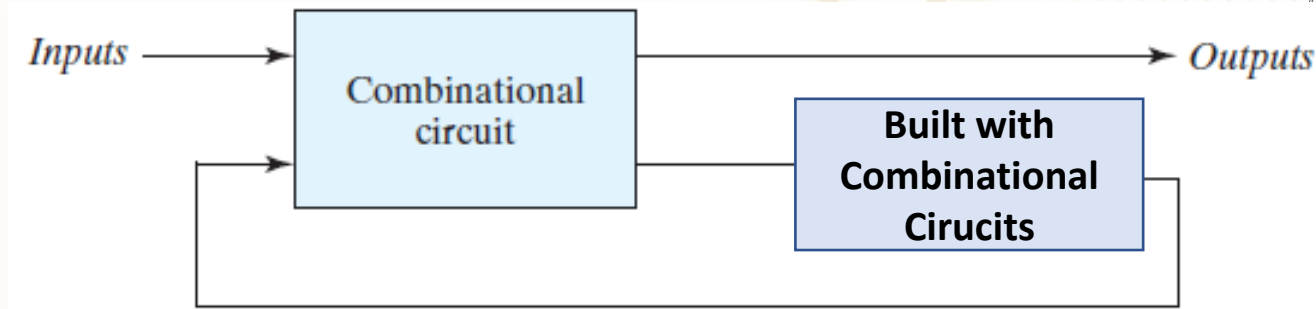


Negative Edge Triggering

Level & Edge Triggered

Edge Triggering	Level Triggering
Edge triggering is based on detecting a sharp edge in the input signal.	Level triggering is based on recognizing a specified signal level.
Edge triggering is frequently used in synchronous circuits, such as counters and flip-flops.	Level triggering is frequently employed in applications that call for continuous monitoring of an input signal, such as data acquisition and control systems.
Edge triggering is particularly helpful in applications that need precise timing.	Level triggering lacks the ability to regulate precise timing.
When edge triggering is used, the output signal is activated when the trigger edge is detected and changes to the opposing state. As long as the input signal is at or above the trigger level.	In level triggering, the output signal will remain in the triggered condition.

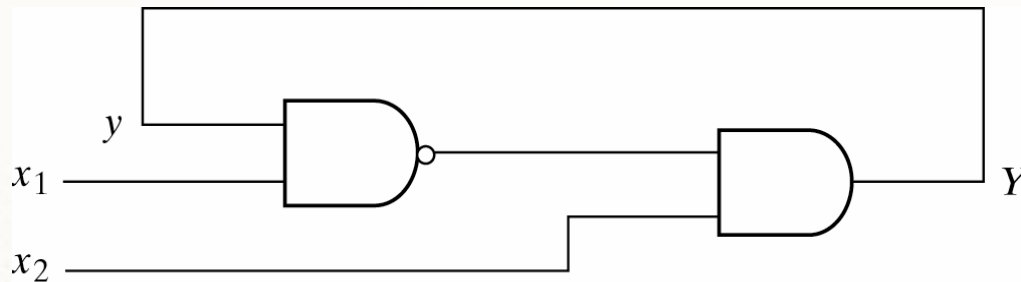
Asynchronous Sequential Circuits



- In Asynchronous Sequential circuits, the **feedback** or **memory** elements are **built with combinational circuits**
- There is a **delay** involved in the **feedback** reaching, as inputs, due to the **propagation delay** of the feedback elements
- There is **no clock signal** here which synchronizes the activities, **leading to unstable state** or **race conditions**
- Due to this, asynchronous circuits are **rarely used**, though it is much **faster than the synchronous sequential circuits**

Example: Asynchronous Sequential Circuits

- Combinational circuits with a feedback circuit
- An asynchronous sequential circuit may become unstable and oscillate between **unstable** states because of the presence of feedback.



The excitation function is:

$$Y = (x_1 y)' x_2 = (x_1' + y') x_2 = x_1' x_2 + x_2 y'$$

and the transition table for the circuit is:

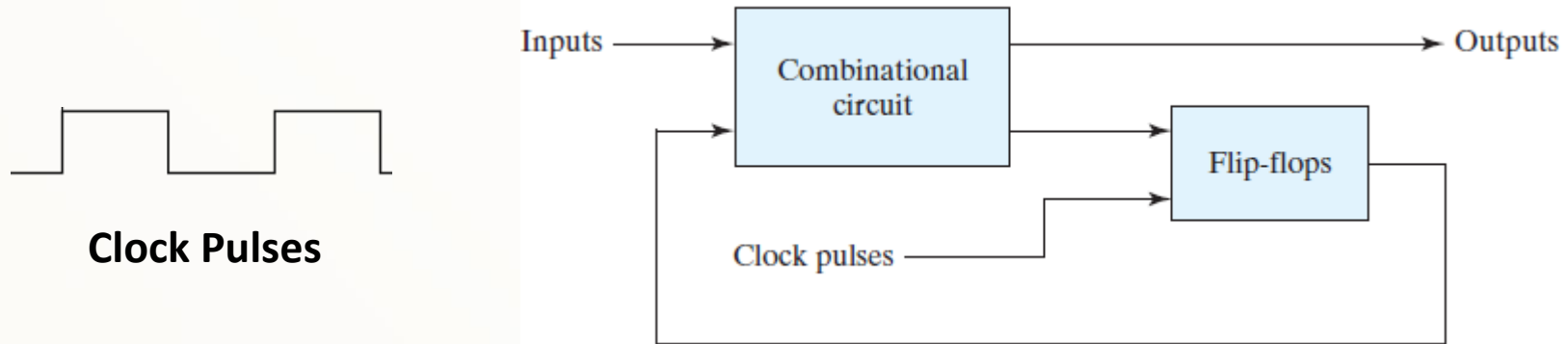
		$x_1 x_2$			
		00	01	11	10
y	0	0	1	1	0
	1	0	1	0	0

Those values of Y that are equal to y are circled and represent stable states. When the input $x_1 x_2$ is 11, the state variable alternates between 0 and 1 indefinitely.

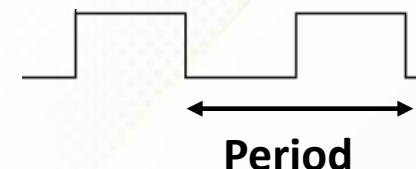
Let us see what happens the inputs x_1 and x_2 are both are 1s.

We can notice that the output Y will alternate between 1 and 0, when both the inputs are 1s.

Synchronous Sequential Circuits



- In **synchronous circuits** the **memory** elements are **clocked flip-flops**
- **Transition** of **states** happen **only during** the **clock edges**
- **State** of the **flip-flops** will be "**frozen**" **until** the **next** (correct) transition of the **clock**
- Each **state** is kept **unaltered** in the **flip-flops** (memory elements) for **one period of the clock**
- Let us study about flip-flops now ..



Storage Elements Latch & Flipflop

Storage Elements – Latches/Flip-flops

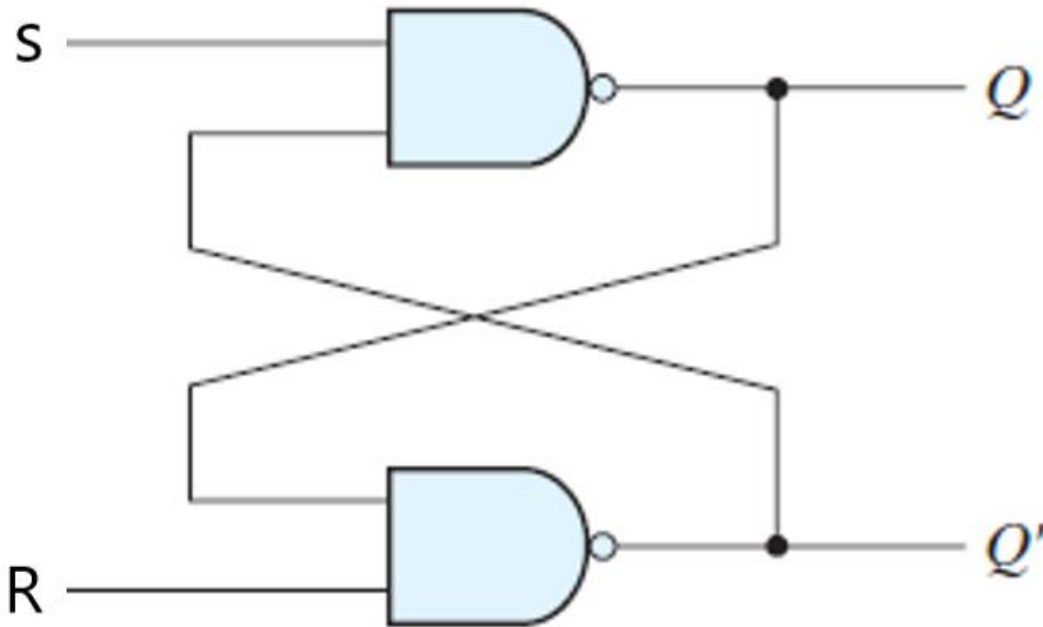
- Storage element in a digital circuit can maintain a binary state indefinitely
 - As long as power is delivered to the circuit
 - And until directed by an input signal to switch states
- Storage elements that operate with **signal levels** (rather than **signal transitions** or **clock edges**) are referred to as **latches**
 - Those **controlled** by a **clock transitions** are **flip-flops**
 - **Latches** are said to be **level sensitive devices**
 - **Flip-flops** are **edge-sensitive devices**

Latch Vs Flipflop

NO	Flip-flop	Latch
1	Flip-flop is a bistable device i.e., it has two stable states that are represented as 0 and 1.	Latch is also a bistable device whose states are also represented as 0 and 1.
2	It checks the inputs but changes the output only at times defined by the clock signal or any other control signal.	It checks the inputs continuously and responds to the changes in inputs immediately.
3	It is a edge triggered device.	It is a level triggered device.
4	They are classified into asynchronous or synchronous flipflops.	There is no such classification in latches.
5	It forms the building blocks of many sequential circuits like counters.	These can be used for the designing of sequential circuits but are not generally preferred.
6	Flip-flop always have a clock signal	Latches doesn't have a clock signal
7	Flip-flop can be build from Latches	Latches can be build from gates
8	ex:D Flip-flop, JK Flip-flop	ex:SR Latch, D Latch

Storage Elements S'R' Latches – With NAND Gates

SR Latch

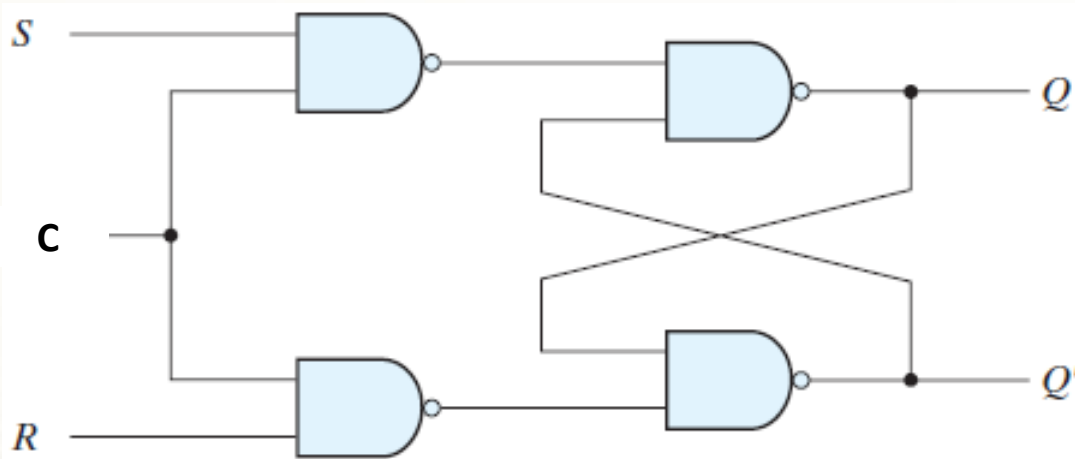


SR Latch Truth table

S	R	Q	Q'
0	0	Invalid	
0	1	1	0
1	0	0	1
1	1	No change	

SR Flipflop

Logic Diagram



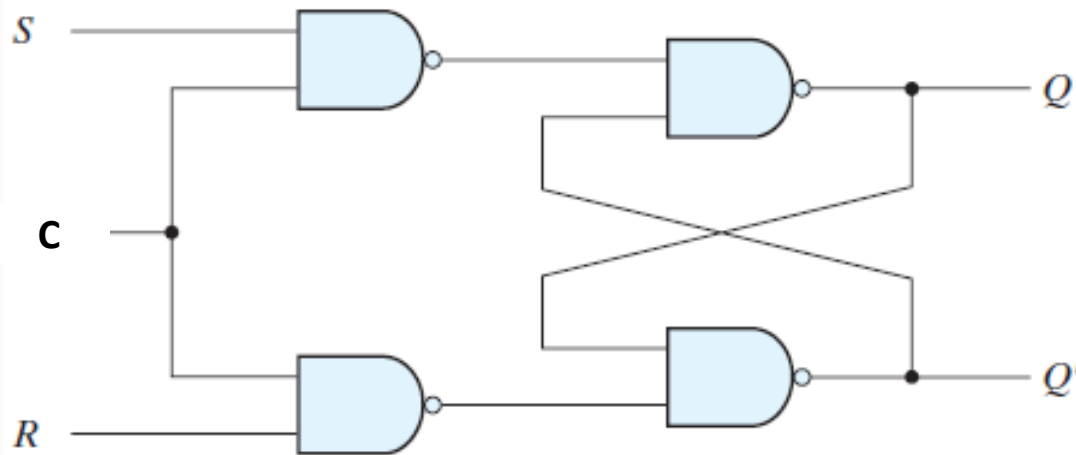
Always **Set** means setting a value to **ONE**
Always **Reset** means resetting a value to **ZERO**

S	R	Q	Q'
0	0	0	1
0	1	0	1
1	0	1	0
1	1	∞	∞

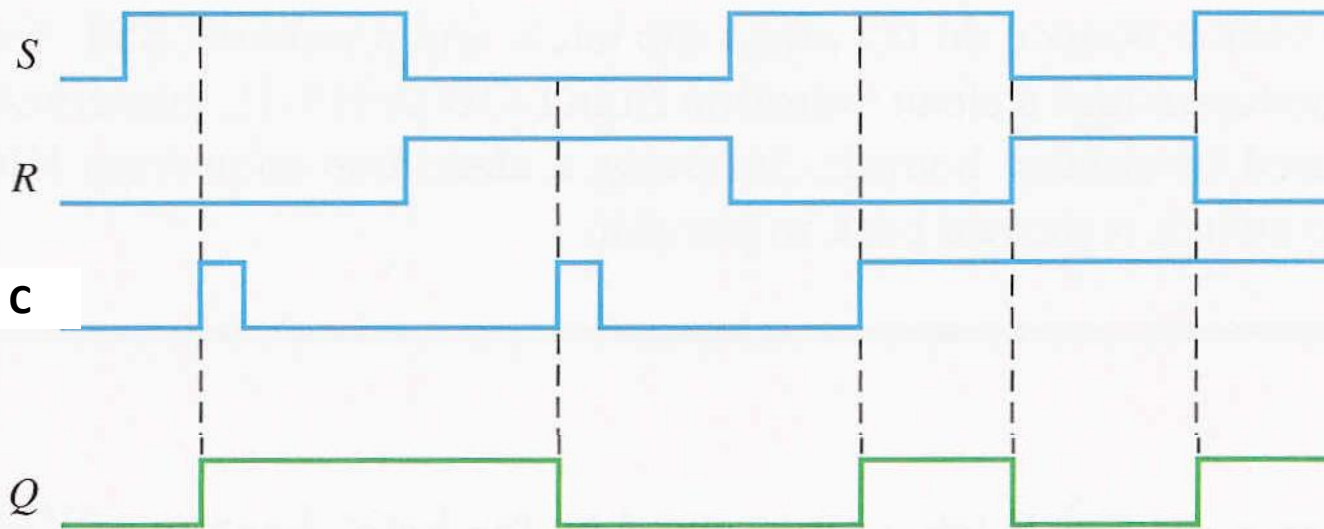
It is forbidden because
If both **Set (1)** and **Reset (1)**
the next state of
Q and Q' are indeterminate

Quiz 1: Draw the output waveform

(Assume Q is initially zero)



C	S	R	Next state of Q
0	X	X	No change
1	0	0	No change
1	0	1	Q = 0; reset state
1	1	0	Q = 1; set state
1	1	1	Indeterminate



ANS:

Session 3.1: Summary

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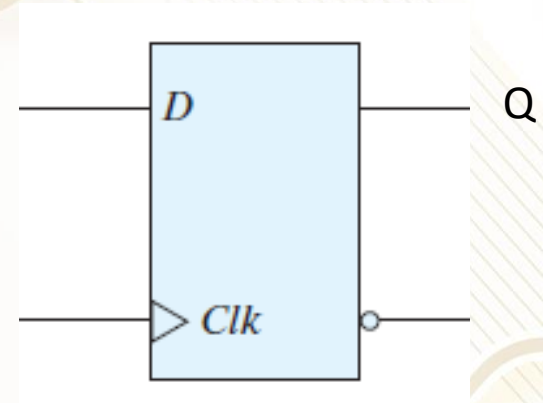
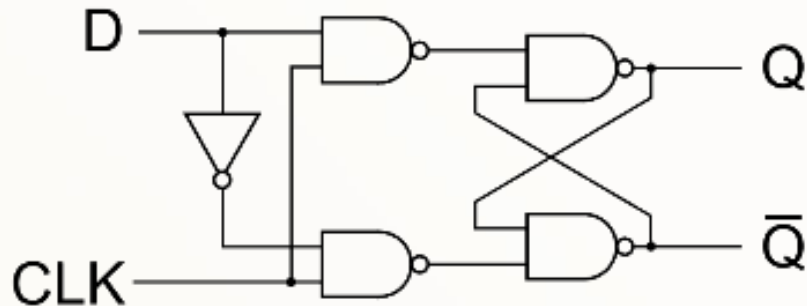
Session 3.2: JK Flip-flops

Session 3.2: Focus

- Latches and Flip-flops
 - Level and Edge-triggered clocks
 - Clock edges (+ve and –ve)
- Edge-triggered D Flip-flops
 - Positive edge-triggered D Flip-flop Implementation
 - Logic Symbol
- JK Flip-flop
 - Implementation

Positive Edge-triggered D Flip-flop

D Flip Flop



D Flip-Flop

D	Q(t + 1)	
0	0	Reset
1	1	Set

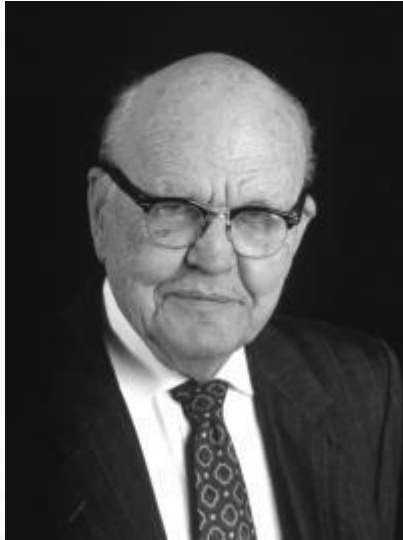
- The value that is produced at the output (**Q**) of the flip-flop is the **value** that was **stored** in the **master** stage **immediately before** the **positive edge occurred**
- The logic symbol of the positive edge-triggered D-flip-flop is give above

JK Flip-flop

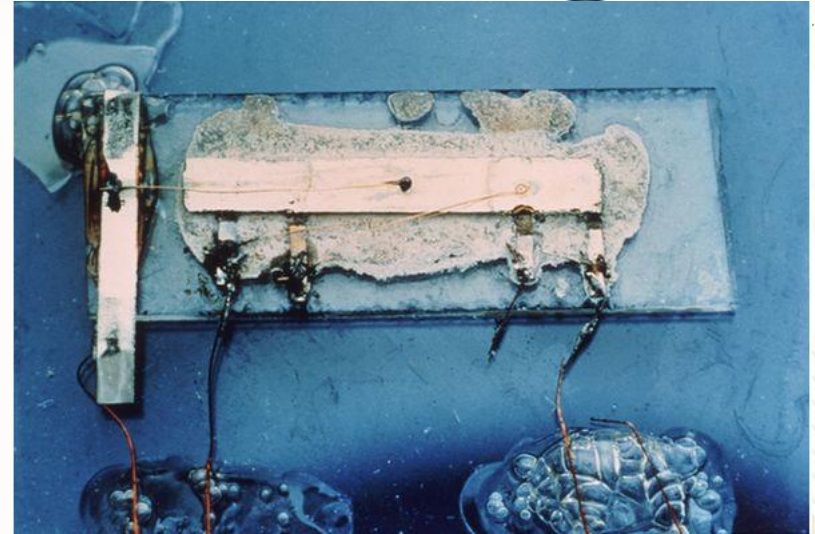
Other Flip-flops with D Flip-flop

- D Flip-flop is the simplest flip-flop that can be built with a less number of logic gates
- In a typical Integrated Chip has millions of logic gates and flip-flops inside them
- Other types of flip-flops are built using the D flip-flop
 - J-K Flip-Flop
 - T Flip-flop

Why is JK Flip-flop named so?

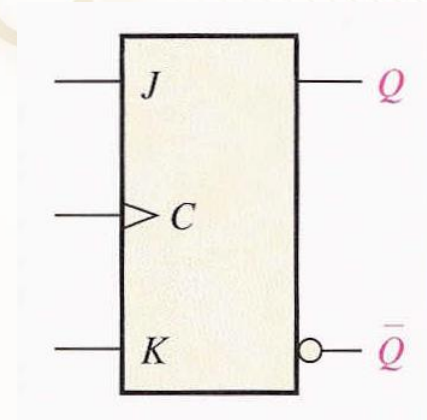
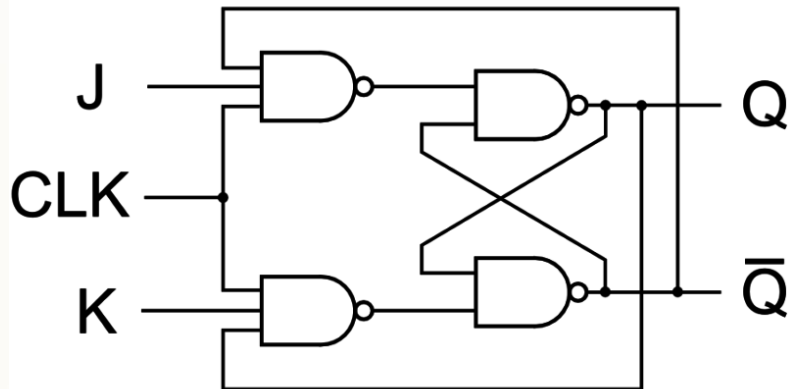


The first
IC built by
JK in 1958



- The choice of the letters "JK" in the **JK flip flop** do have significance.
- The **JK flip flop** was **named** after **Jack Kilby**, the **Texas Instruments (TI)** engineer that **invented** the **integrated circuit (IC)** in **1958**. Won **Noble prize** in **physics** in **2000**.
- The **modified RS circuit** that **eliminated race conditions** was named **JK** in his honour.

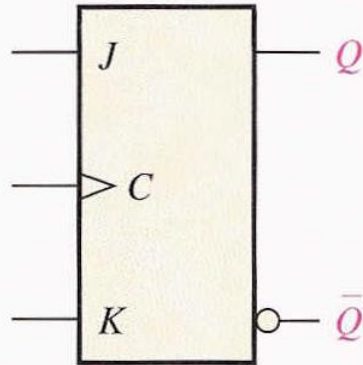
JK Flipflop



JK Flip-Flop

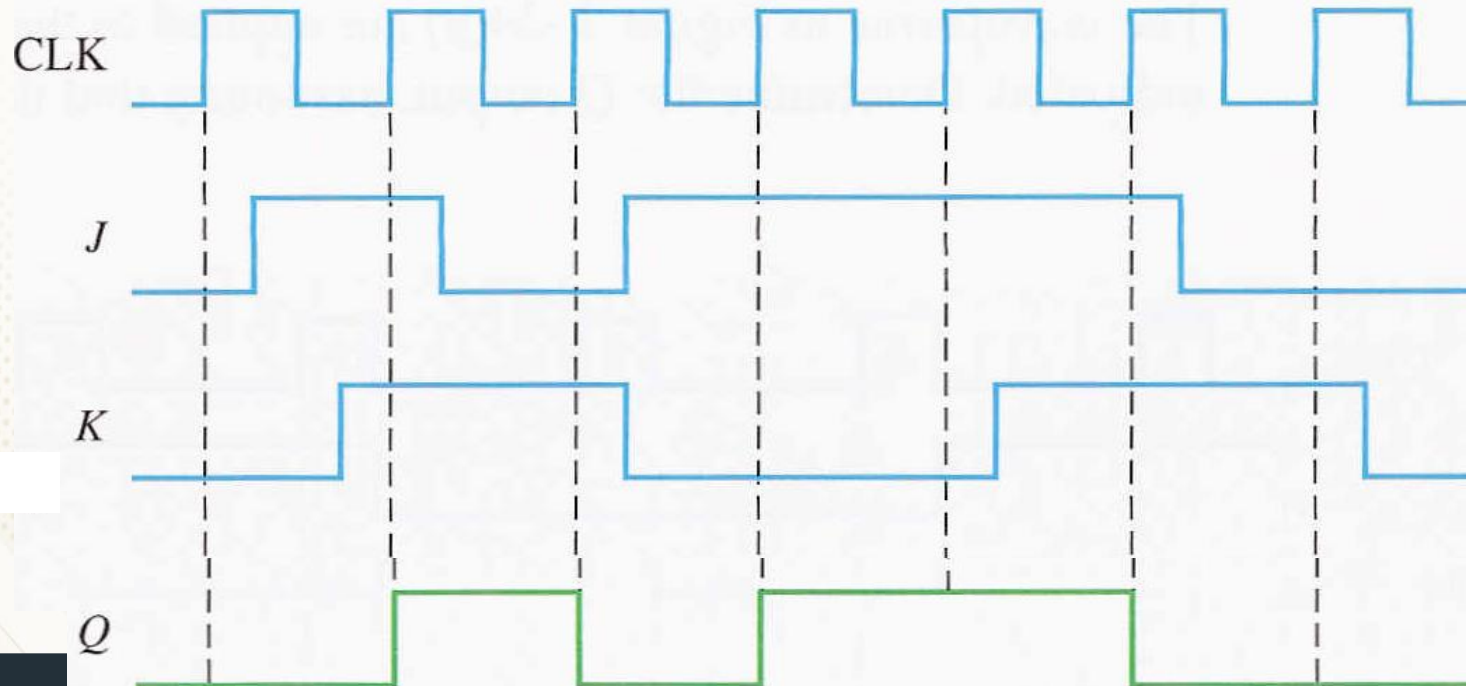
<i>J</i>	<i>K</i>	<i>Q</i> (<i>t</i> + 1)	
0	0	<i>Q</i> (<i>t</i>)	No change
0	1	0	Reset
1	0	1	Set
1	1	<i>Q</i> '(<i>t</i>)	Complement

Quiz 1: Draw the output (Q) waveform

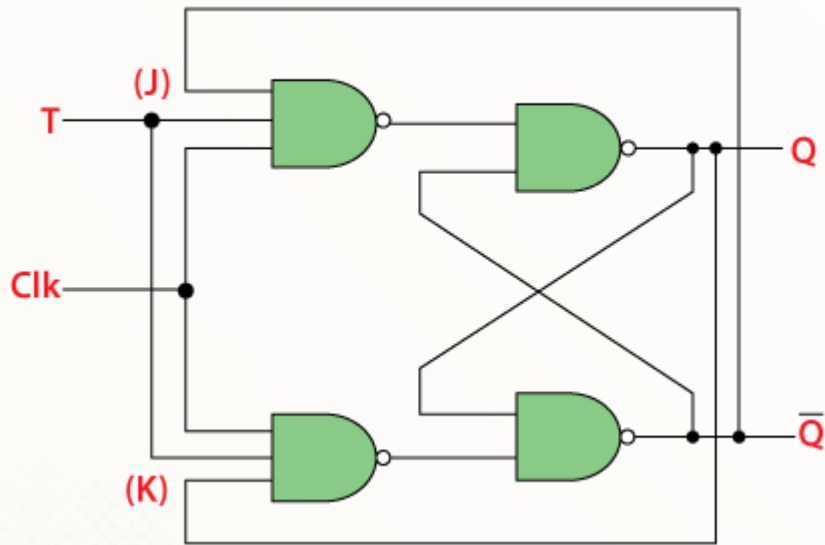


Assume that the flip-flop
is initially **RESET**

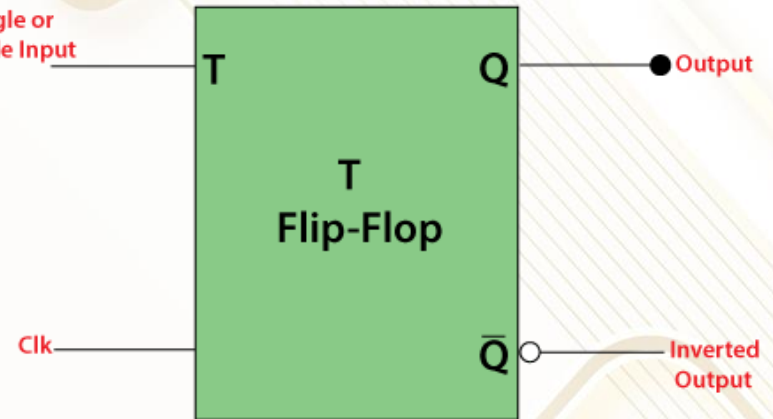
This is **Positive** edge-triggered



T flip flop



Toggle or
Triggle Input



T	Previous		Next	
	Q	Q'	Q	Q'
0	0	1	0	1
0	1	0	1	0
1	0	1	1	0
1	1	0	0	1

Session 3.2: Summary

- Latches and Flip-flops
 - Level and Edge-triggered clocks
 - Clock edges (+ve and –ve)
- Edge-triggered D Flip-flops
 - Positive edge-triggered D Flip-flop Implementation
 - Logic Symbol
- JK Flip-flop
- T fliflop

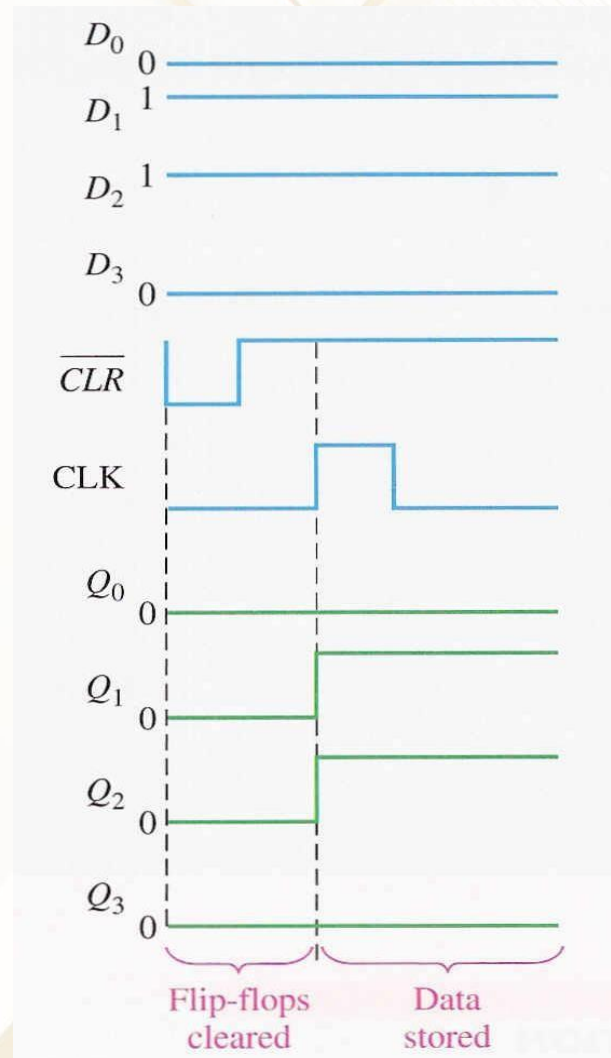
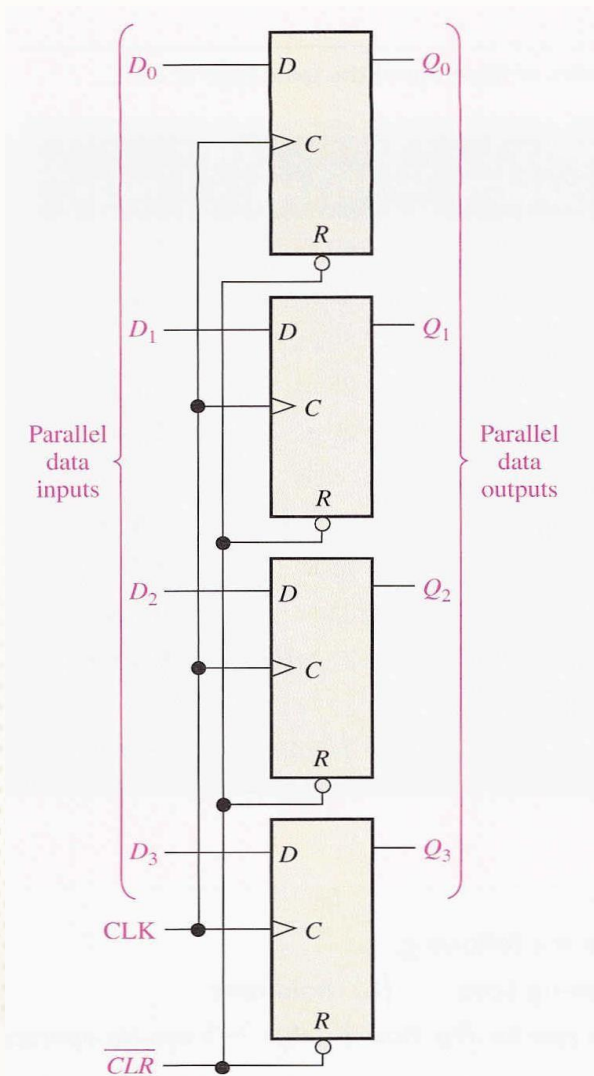
Session 3.3: JK and T Flip-flops

Session 3.3: Focus

- Applications using D Flip-flop
 - 4-bit Asynchronous Up counter
 - 4-bit Asynchronous Down counter
 - Serial to Parallel Data Converter

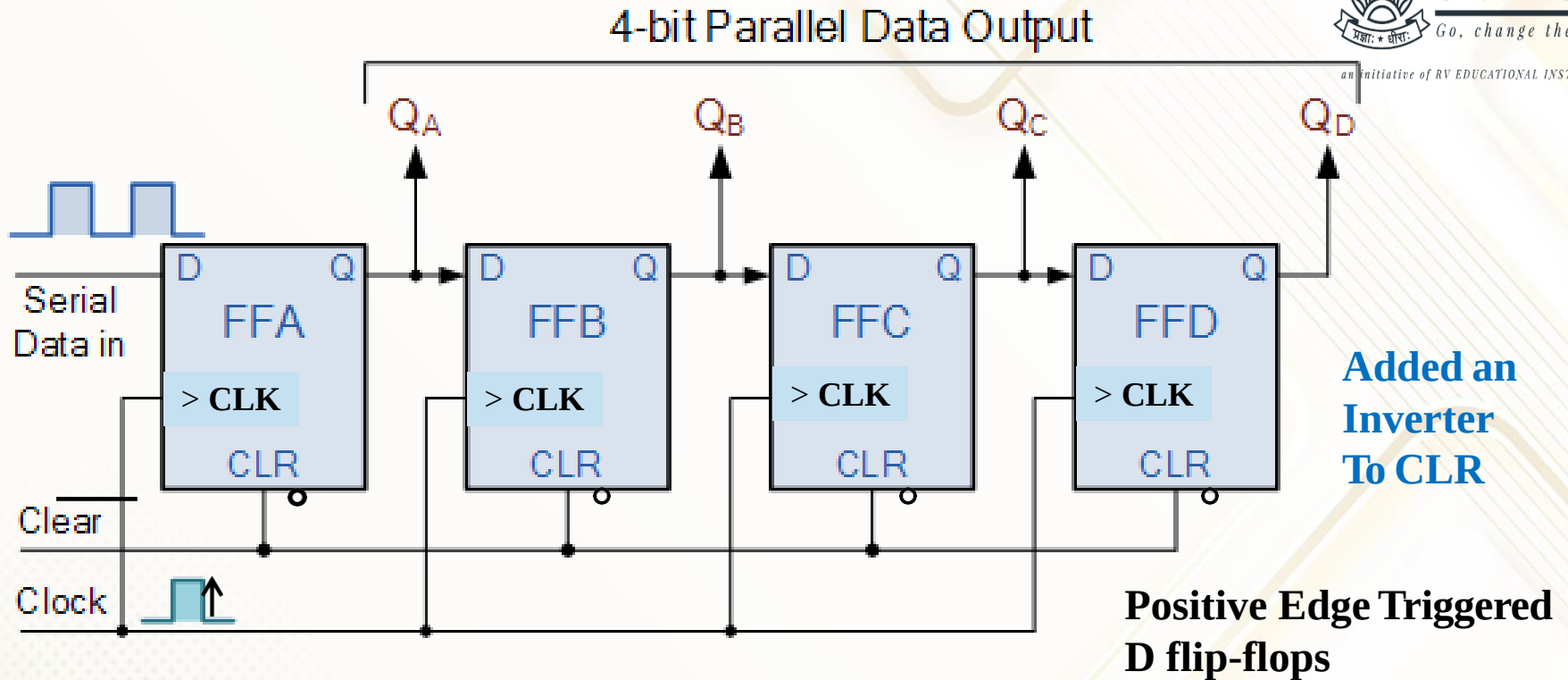
Applications using D- Flip-flops

Quiz 1: What does this circuit do?



**4-bit
Data
storage**

Example 3: Serial to Parallel Data Converter



- Serial to parallel data converter
- The serial data is shifted in at the positive edge of every clock pulse
- After four clock pulses the parallel data is available at $Q_3Q_2Q_1Q_0$

Session 3.3: Summary

- Applications using D Flip-flop
 - 4-bit Asynchronous Up counter
 - 4-bit Asynchronous Down counter
 - Serial to Parallel Data Converter

Session 3.4: FSM, Moore and Mealy Models

Session 3.4: Focus

- Finite State Machine (FSM)
 - Example: Turnstile
- Types of Clocked Sequential Circuits
 - Moore Model
 - Mealy Model
 - Example circuits

Finite State Machine

Example: Turnstile

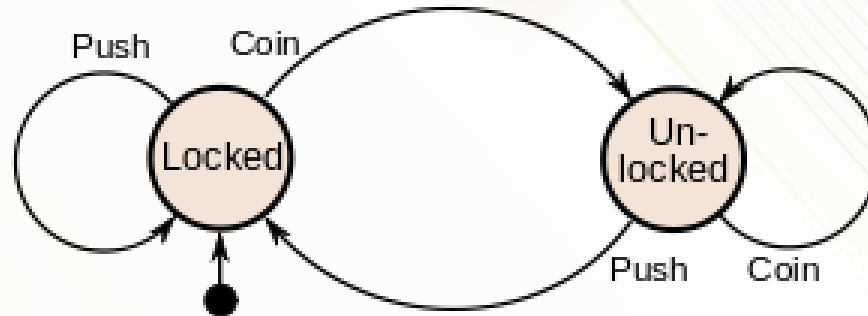


- Allows people to **enter** through **only on one side**. It can also have **controlled access** (coin) to let people in.

Operation of Turnstile Operation Explained

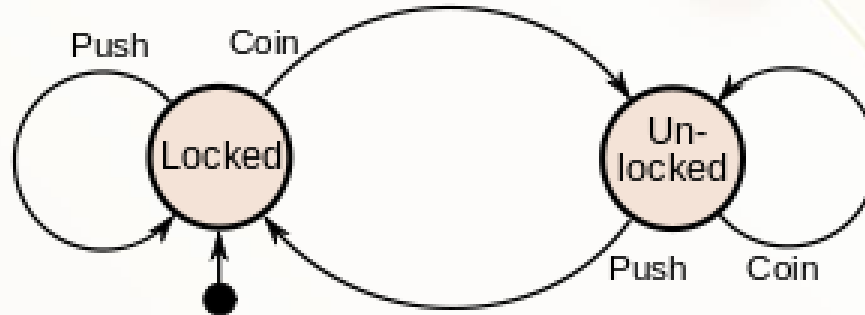
Inputs are:

1. Coin
2. Push



- Operation is modeled with a **State Machine or State Diagram**
- **Turnstile** can be in **one** of these **two** states
 - **Locked**
 - **Unlocked**
- **Initially** it is in a **Locked** condition
- **Dropping** a coin changes the state of the machine from **Locked** to **Unlocked**

Turnstile Operation Explained



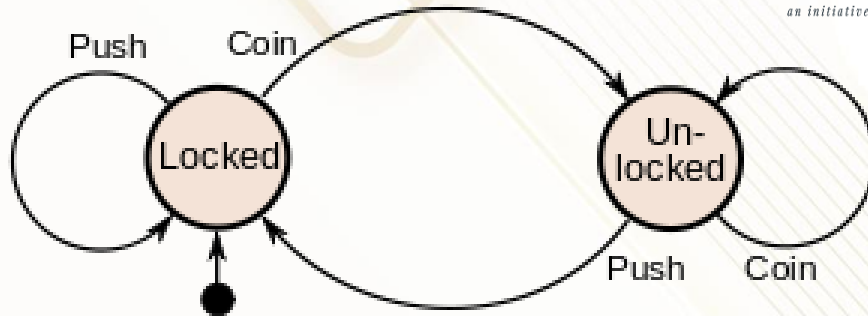
- In the unlocked state, when the cross bar is pushed
 - **One person** can **move in** through the gate
 - Turnstile **changes** back to **Locked state**
 - The cross bar cannot be moved in the locked state even if it gets pushed
- There by **allowing** only maximum **one person** to enter for every **coin dropped**

Finite State Machine (FSM)

- A Finite-State Machine (**FSM**) or Finite-state Automaton (**FSA**, plural: automata), finite automaton, or simply a **state machine**, is a **mathematical model** of computation.
- It is an **abstract machine** that can be in exactly **one** of a **finite** number of **states** at any given **time**.
- The **FSM** can **change** from **one state** to **another** in **response** to some **external inputs**; the change from one state to another is called a **transition**.
- A **FSM** is defined by a **list** of its **states**, its **initial state**, and the **conditions** for **each transition**.

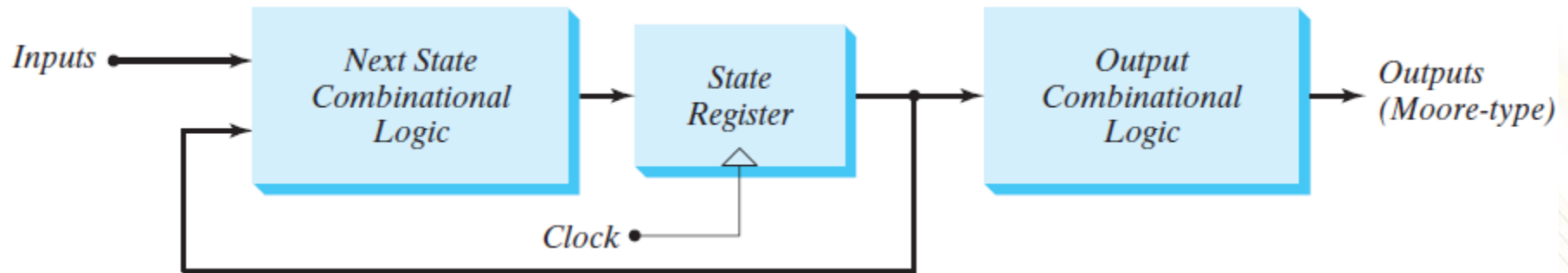
FSM of Turnstile

Dark dot here
Shows the
Initial state



- **List of states**
 - Locked
 - Unlocked
- **Initial State:** Locked
- **External Inputs:** Coin and Push
- **Transitions:**
 - Locked to Unlocked (Condition: Dropping of coin)
 - Unlocked to Locked (Condition: Pushing the bar)

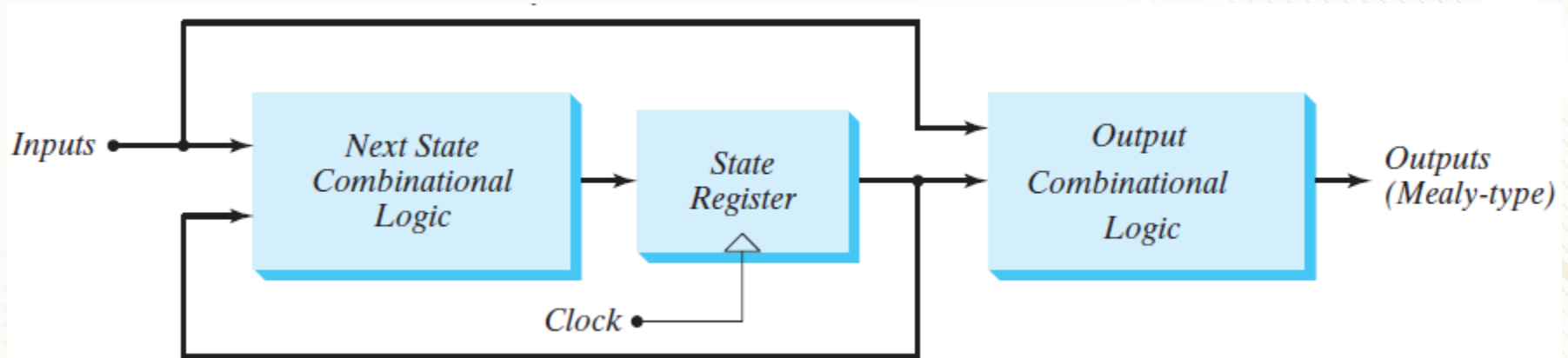
a) Moore Model



- In the **Moore model**, the **output** is a function of **only** the **present state**.
 - Present states are in turn driven by the inputs

b) Mealy Model

- In the **Mealy model**, the **output** is a function of **both** the **present state** and the **inputs**.

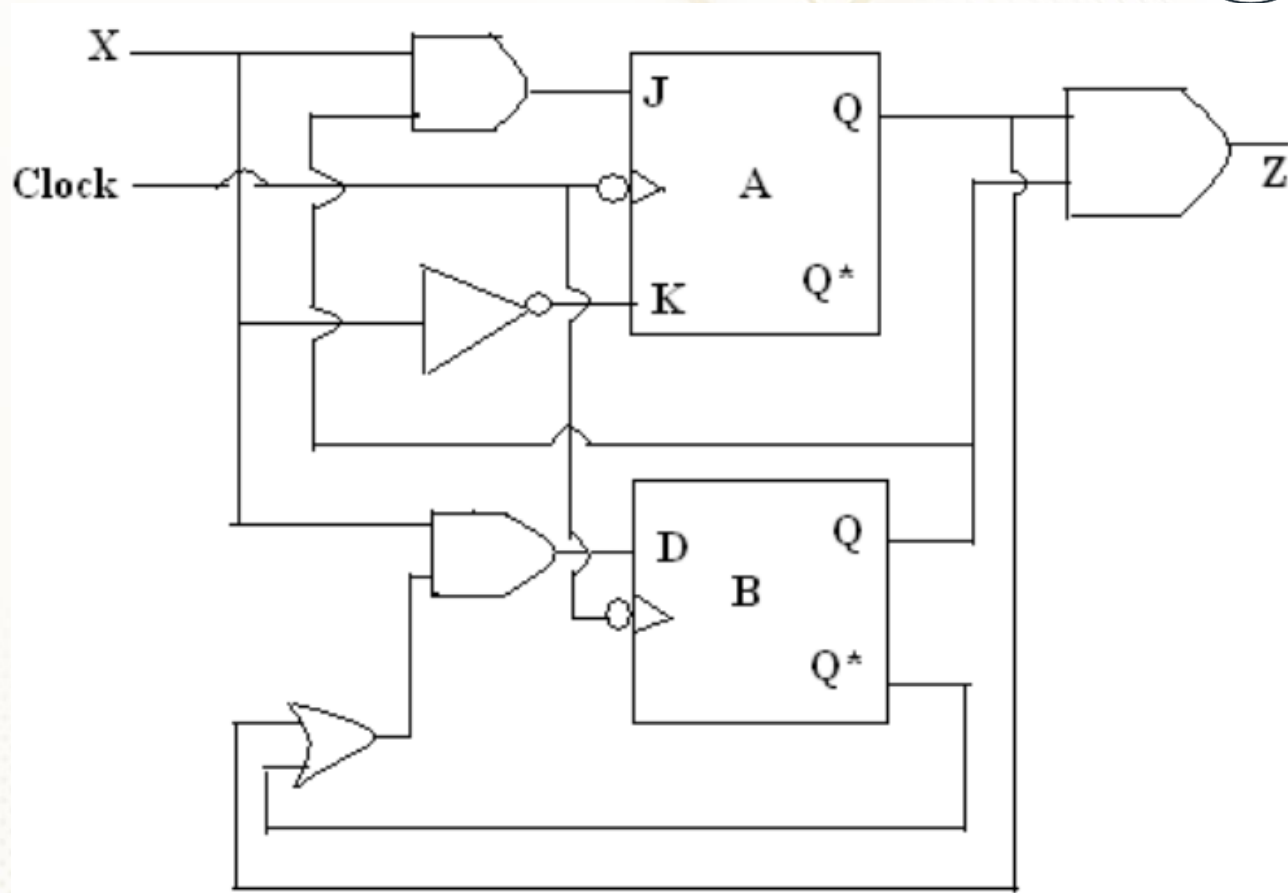


- A circuit may also have both types of outputs.
 - i.e., both Mealy and Moore type of outputs

Mealy and Moore Machines

- These **two models of sequential circuits** are commonly referred to as a **Finite State Machine**, abbreviated **FSM**.
- The Mealy model of a sequential circuit is referred to as a **Mealy FSM** or **Mealy machine**.
- The Moore model is referred to as a **Moore FSM** or **Moore machine**.

a) Moore Model



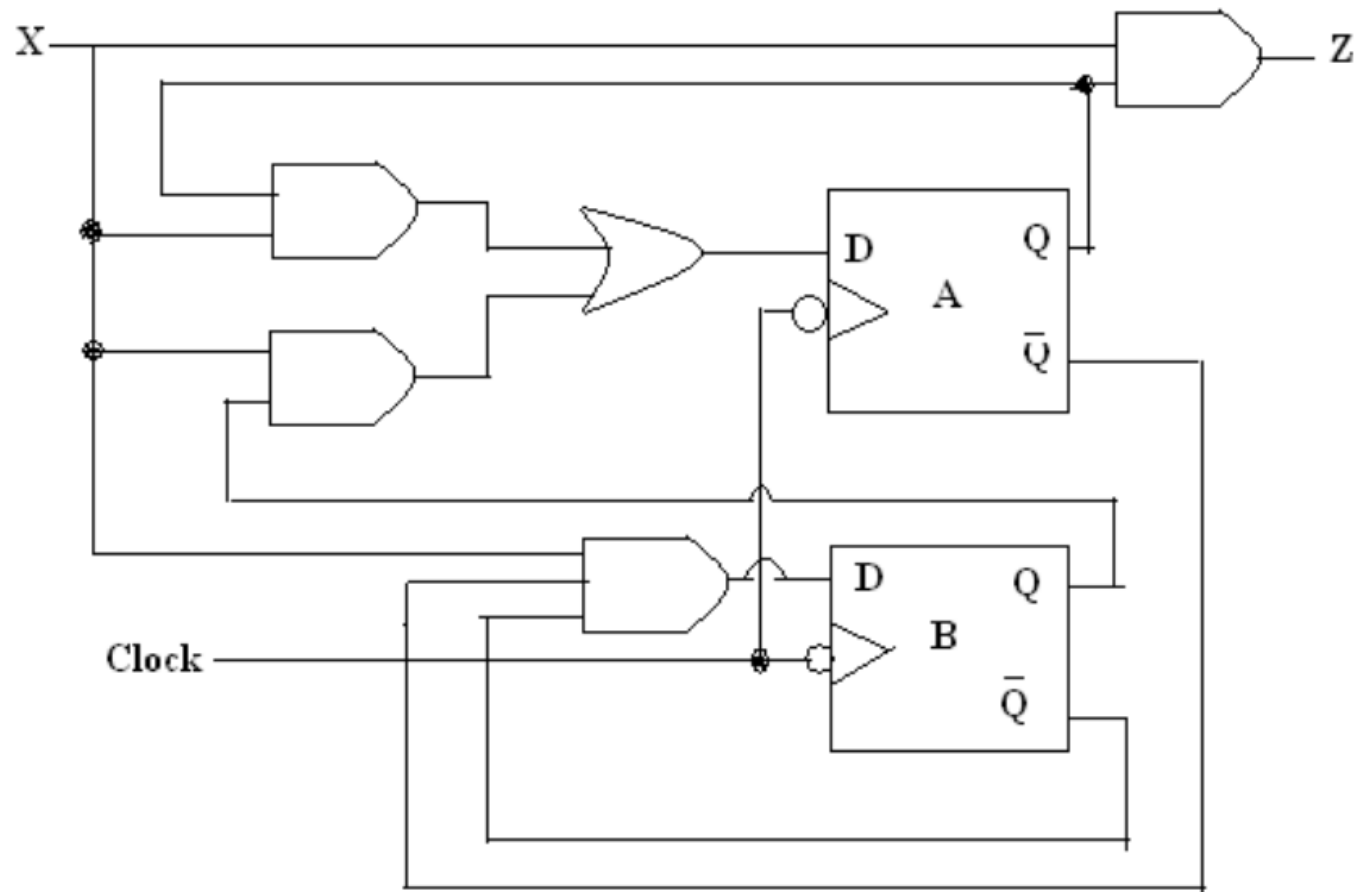
b) Mealy Model



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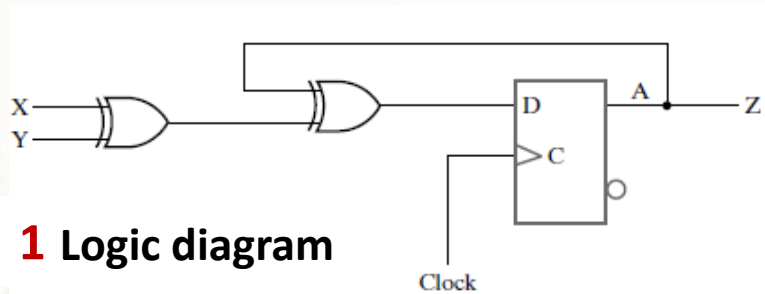
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Steps in the Analysis

1. Circuit/Logic Diagram
2. State Equations
3. State Table
4. State Diagram



$$A(t + 1) = A \oplus x \oplus y$$

2 State equation

3 State table

Present state	Inputs		Next state
A	x	y	A
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

